Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Core Pure 1 – Linear Transformations

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Write the matrix for a reflection in the x-axis. **[2]**

2. Write the matrix for a reflection in the y-axis. **[2]**

3. Write the matrix for a rotation of $90°$ anticlockwise about the origin. **[2]**

4. Write the matrix for an enlargement scale factor k centre origin. **[2]**

5. Find the image of the point $(2, 1)$ under reflection in the x-axis. **[2]**

6. Find the image of $(1, 0)$ under rotation $90°$ anticlockwise. **[2]**

7. Describe the transformation given by $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. **[2]**

8. Describe the transformation given by $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$. **[2]**

9. Find the matrix for reflection in the line $y = x$. **[2]**

10. Find the matrix for rotation of $180°$ about the origin. **[2]**

11. The matrix $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ represents a shear. Describe it. **[3]**

12. Find the image of the unit square under $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$. **[3]**

13. State the area scale factor of the transformation with matrix A, in terms of $\det A$. **[2]**

14. Find the matrix for rotation of $45°$ anticlockwise. **[3]**

15. Describe the transformation $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$. **[2]**

Section B – Hard (Q16–25)

16. Find the single matrix for reflection in the x-axis followed by rotation $90°$ anticlockwise. **[4]**

17. A transformation has matrix $\begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$. Find the image of the line $y = x$. **[4]**

18. Find the invariant points of the transformation $\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$. **[4]**

19. Find the invariant lines of $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$. **[4]**

20. The matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ is applied twice. Describe the resulting transformation. **[3]**

21. Find a matrix that represents a stretch scale factor 2 parallel to the x-axis. **[2]**

22. Show that the transformation with matrix $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ maps the line $x = 1$ to itself (invariant line). **[4]**

23. A composite transformation is rotation $90°$ then reflection in $y = x$. Find the matrix and describe the single equivalent transformation. **[5]**

24. Find the area of the image of a triangle of area 5 under the matrix $\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$. **[3]**

25. Determine whether $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ represents a transformation that is invertible, and interpret geometrically. **[4]**

End of Worksheet • *Show all working clearly.*